The Paleomagnetism of the Turkana Lavas[1]

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Abstract. Directions of magnetizations obtained after cleaning the Miocene lavas from northwest Kenya are presented. The pole position thus obtained, together with similar data from eastern Africa, indicates that no significant change has occurred from late Eocene to recent times. Observed directions suggest that a late Miocene reversed epoch has occurred.

Introduction. As part of a paleomagnetic study of the Rift Valley volcanics of eastern Africa, 65 oriented samples of lavas from the Turkana area of northwest Kenya were collected in August 1963. These were collected either as blocks or as cores using a portable drill similar to that of *Graham and Keiller* [1960]. The directions of the natural remanent

[1] Abstract from a thesis to be submitted by P. K. S. Raja in partial fulfillment of the requirements for the Ph.D. degree of the University of London.

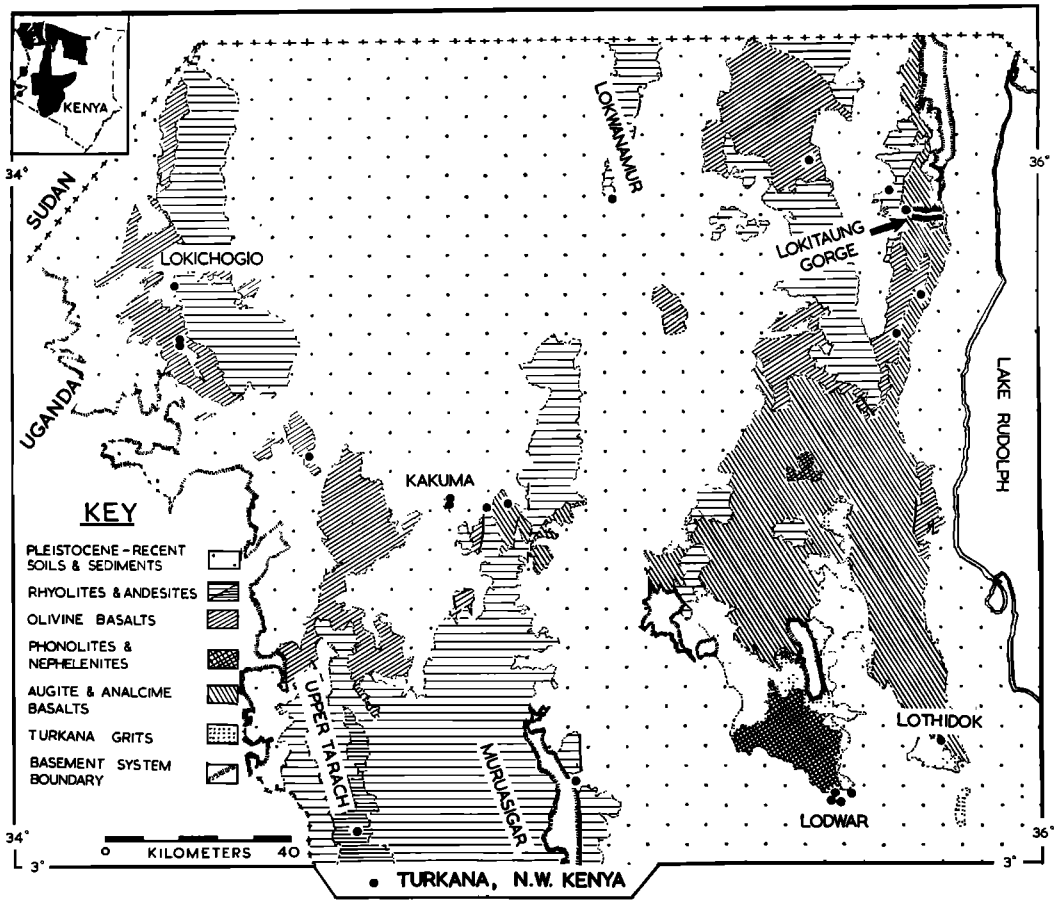


Fig. 1. Geological sketch map of the Turkana area of northwest Kenya. The Rift volcanics are indicated by solid shading in the key.

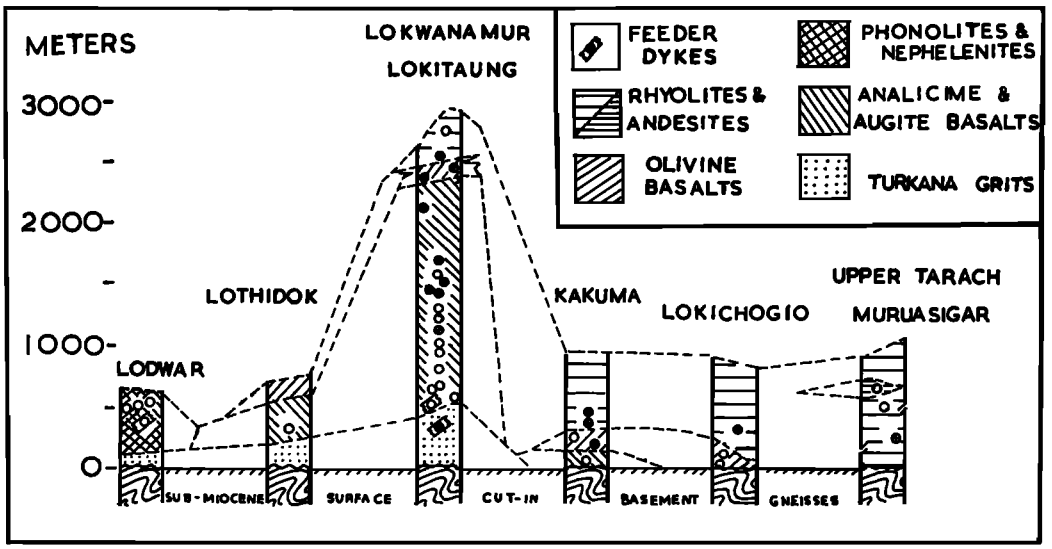


Fig. 2. Diagramaatic stratigraphic section of the Turkana area.
● Sites with normal polarity.
○ Sites with reversed polarity.
⊕ Sites that could not be cleaned. Polarity unreliable.

magnetizations of specimens 22 mm in diameter and 25 mm long, taken from these samples, showed a high degree of scatter when measured under an astatic magnetometer. Preliminary results, based on a scatter criterion for the selection of samples, have been reported by *Mussett et al.* [1964].

Geology of the area. Figure 1 indicates the major stratigraphic units of the Turkana area and shows the sampling sites. The relationship between these units and the vertical distribution of sampling is shown by the diagrammatic sections of Figure 2, in which the sub-Miocene surface has been restored to horizontal to remove the complications of rift faulting. The augite and analcime lower basalts were extruded after the deposition of the Miocene Turkana grits to the east. The olivine upper basalts represent a basic basaltic phase only locally developed. This was probably preceded by extrusion of the phonolites and nephelenites of the Lodwar area, although no contact between them and the upper basalts is seen. Intercalations of rhyolites with basaltic lavas east of Lokwanamur suggest that at the start of rhyolite emplacement, both upper and lower basalts were being extruded at different localities.

Within each stratigraphic unit there is much variation in composition, grain size, vesicularity, etc. Severe weathering and mechanical breakdown is common in the lower basalts and

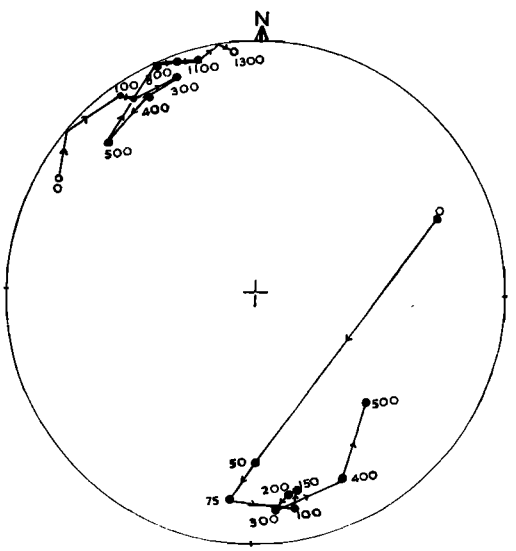


Fig. 3. Polar projection showing changes in direction of magnetization during alternating field demagnetization.
● Dips down. ○ Dips up.

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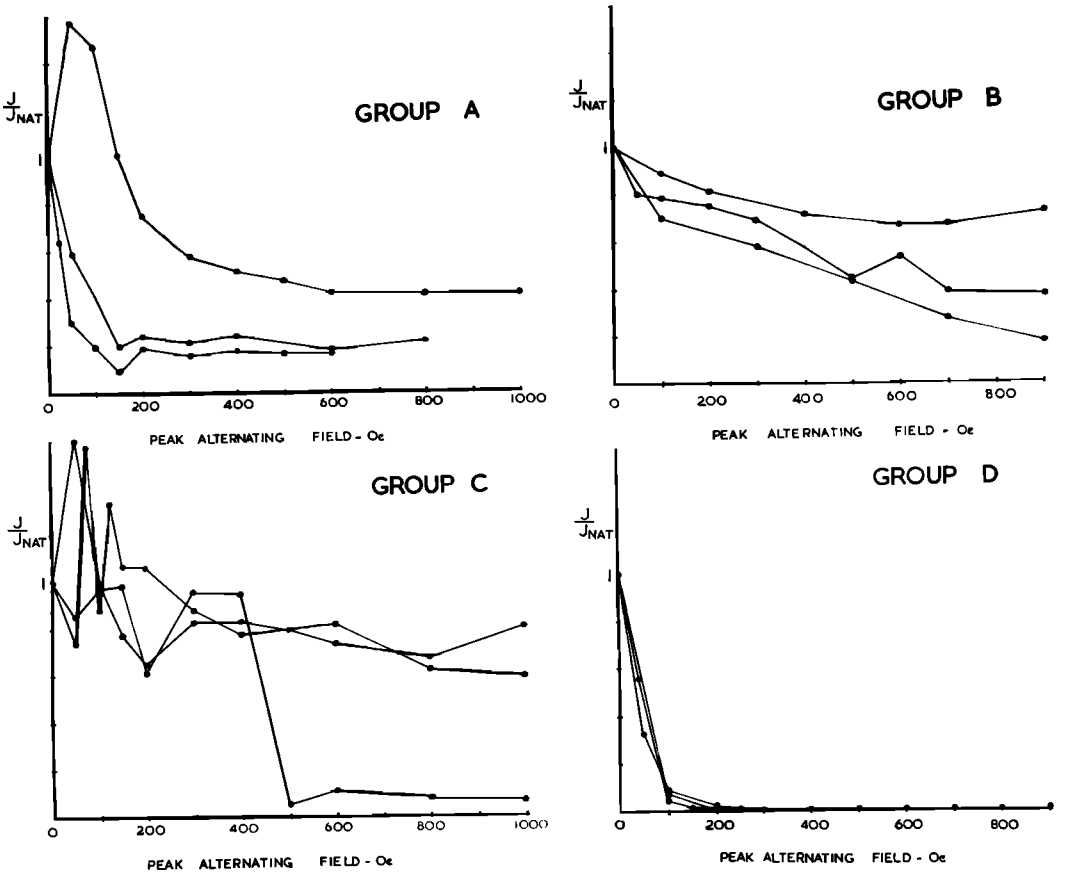


Fig. 4. Alternating field demagnetization curves. Intensities, J, are normalized.

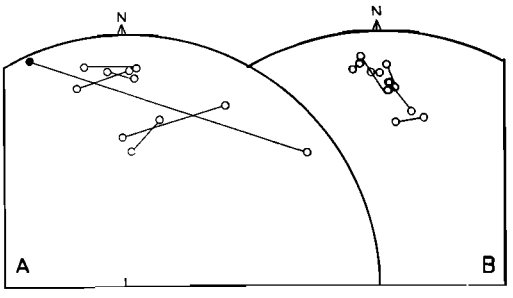


Fig. 5. Comparison of thermal and alternating field demagnetization.
A. Solid lines link specimens from the same sample before cleaning.
B. Thermally cleaned specimens are linked with specimens from the same sample cleaned by alternating field demagnetization.

● Dips down. ○ Dips up.

occurs to a lesser degree in the other units. The occurrence of secondary minerals such as iron ores, quartz, calcite, uralite, and chlorite is common, and the glassy groundmass of many of the rhyolites is often devitrified. Polished sections have not been examined in detail. Feeder dikes cutting all units are common, and four of these were sampled. No suitable baked contacts were observed. Intercalated pyroclastics and sediments occur most often lower in the succession.

Alternating field demagnetization. Remeasurement of the natural remanent magnetization of specimens after storage of more than 1 year gave different directions from those formerly observed, which indicates that the specimens had acquired viscous magnetization during storage. *Nairn* [1964] has observed that Miocene lavas collected from the Nairobi area behaved

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similarly. Secondary magnetizations were subsequently removed by progressive alternating field demagnetization, the method employed being similar to the one described by *McElhinny and Gough* [1963]. We demagnetized at least one specimen from each site in fields progressively increased up to 1300 oe. The field at which the directions of magnetization remained constant while the intensity of magnetization decreased was chosen as the demagnetizing field for all the other specimens from the site.

Figure 3 shows typical changes in direction of magnetization in two specimens during progressive demagnetization. The first specimen (rhyolite) required a peak field of nearly 1300 oe to reach a stable direction, whereas the second (basalt) needed only 150 oe. The direction of magnetization of this specimen then showed little change below 300 oe but became scattered in higher fields.

Altogether 128 specimens from 65 samples collected from 35 sites were demagnetized as described above. When the normalized intensities of the demagnetized specimens were plotted against the demagnetizing fields, the curves were found to fall roughly into four groups as shown in Figure 4.

The curves of group A shows an initial rapid change of intensity in low fields, thereafter showing little change. This is attributed to a soft secondary component which may make an arbitrary angle with the stable component. The group consists mainly of basalts that could be cleaned in fields of about 200 oe.

The curves of group B do not show initial rapid changes of intensity, which presumably indicates the absence of a soft secondary component. This group contains only rhyolites, which could be cleaned in high fields as illustrated in Figure 3.

The curves of group C are characterized by fluctuations in intensity. These specimens showed irregular changes in direction of magnetization, and they could not be cleaned.

The curves of group D fall very rapidly to less than 1% of their initial intensity in low demagnetizing fields. The following observations suggest that the specimens of this group might have acquired isothermal remanent magnetization from lightning current:

1. High itensities: $> 100 \times 10^{-4}$ emu/g.
2. Typically steep or shallow dips and scattered directions (due to vertical or horizontal currents?).
3. Although the surface specimen of one core could not be cleaned, the bottom ones could be cleaned in high field.
4. Nearby samples showed normal intensities, and they could be cleaned in low fields.

Similar effects of lightning were noticed by *McElhinny and Gough* [1963] and *Graham* [1961].

Thermal demagnetization. Some specimens

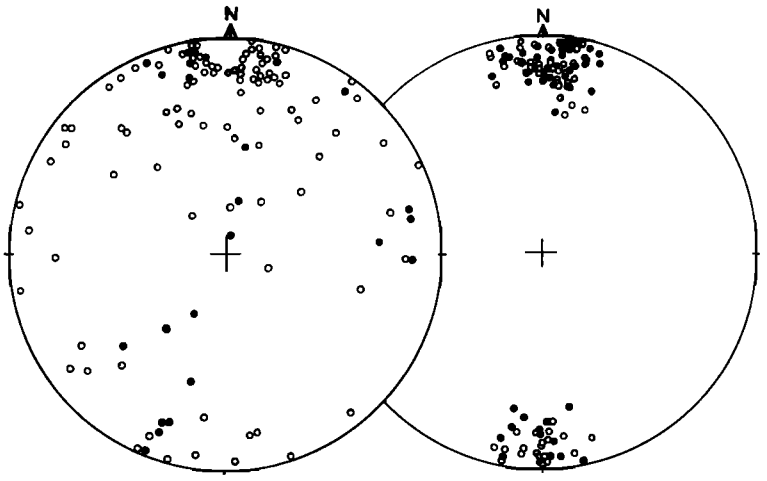


Fig. 6. Polar projection showing directions of magnetization before and after cleaning. ● Dips down. ○ Dips up.

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TABLE 1. Summary of Paleomagnetic Results for Tertiary Volcanics of Eastern Africa

Age, Locality, and Reference	Collection Data						Directions of Magnetization					Pole Positions		
	F	St	Sm	Sp	S. W.	Polarity	D°	I°	K	α°	Cleaning	Lat°N	Long°E	α^1/α^2
Pleistocene (Nakuru, Kenya) *Nairn* [1964]	4	5	11	27	Sm	N + R	[11.9	−11.3	40	7.3]*	Af	[76.9	207	3.8/7.4]
Pleistocene (Olduvai, Tanzania) *Grommé and Hay* [1964]	4	1	4	12	Sp	N	4.8	+1.6	220	0.9	Af + Th	84.7	100	[0.45/0.9]
Plio-Pleistocene (Rungwe, Tanzania) *Nairn* [1964]	4	3	9	32	Sm	N + R	[16.2	−28.6	22	11.3]	Af	73	214	6.8/12.4
Plio-Pleistocene (South Rift, Kenya) *Mussett et al.* [1964]	22	35	83	167	Sp	N + R	358.9	+6.6	9	5.1	Sr	86	203	2.6/5.1
Mio-Pliocene (Nairobi, Kenya) *Nairn* [1964]	6					N + R	7.6	+5.1	46	9.9	Af	82	139	[4.9/9.9]
Miocene (Turkana, Kenya) *Mussett et al.* [1964]	4	11	23	54	Sm	N + R	2.1	−8.5	22	6.6	Sr	81	202	3.4/6.7
Miocene (Turkana, Kenya) Present paper	4	35	57	108	Sm	N + R	2	−1	30	3	Af + Th	85	193	1.5/3
Eocene (Ethiopia) *Grasty* [1965]	7	11	21	42	Sm	N	3	+14		7	Th	86	86	3.7/7.7

F, number of flow units included in calculating results.
St, number of sites included in calculating results.
Sm, number of samples included in calculating results.
Sp, number of specimens included in calculating results.
$\alpha^\circ/\alpha^1/\alpha^2$, semiaxis/semiaxes of cone/oval of 95% confidence.
Af, cleaning by alternating field demagnetization.
Th, cleaning by thermal demagnetization.
Sr, site rejection based on scatter of NRM's.
S. W., statistical weighting.
N, normal polarity.
R, reversed polarity.
D°, declination in degrees.
I°, inclination in degrees (positive down).
K, dispersion parameter [*Fisher*, 1953].
[ ]*, calculated from published results.

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were also thermally demagnetized beneath an astatic magnetometer and their high-temperature directions deduced from vector heating diagrams as described by *Wilson* [1961]. Figure 5 compares the results obtained by thermal demagnetization with those obtained by alternating field demagnetization, and the close agreement shows that both methods are equally acceptable.

Results of demagnetization. In Figure 6 the directions of magnetization for all specimens before cleaning are compared with the directions after cleaning for all specimens that could be cleaned.

The mean direction of magnetization and statistical data [*Fisher,* 1953] after cleaning is given in Table 1, and the results, based only on selected natural remanent magnetizations [*Mussett et al.,* 1964] agreed with it. The means were calculated after reversing the directions of the reversed specimens. Dips due to block faulting are low and do not significantly affect mean directions of magnetization. Mean directions calculated for each of the main stratigraphic units agreed within their circles of confidence.

Polarity changes. Cox et al. [1964] have investigated polarity changes back to 4 m.y. ago and have concluded that these changes are systematic, a given polarity being dominant over a period of the order of 1 m.y. Such polarity 'epochs' may be interrupted by short switches of polarity, which they have termed 'events.' The distribution of the site polarities has been indicated in Figure 2. These polarities may be due to self or field reversal. It can be seen that reverse polarities are dominant over the period from the middle of the lower basalts to the beginning of rhyolite extrusion. It could tentatively be suggested that this represents a reversed epoch, during which the few normal sites may represent normal events. Further sampling is needed to test this working hypothesis.

Note added in proof. K-Ar datings for samples of the upper and lower basalts near Lokitaung and the phonolites and nephelenites near Lodwar have been recently carried out by R. L. Grasty and J. A. Miller of the Department of Geodesy and Geophysics of Cambridge University, England. These indicate an upper Oligocene to lower Miocene age for the basalts and a mid-Miocene age for the phonolites and Nephelenites. Details and discussion of these results will be published in the near future.

Acknowledgments. We wish to thank the Commissioner (Mr. B. H. Baker), Mr. J. Walsh, and Mr. R. G. Dodson of the Department of Mines and Geology, Kenya, for access to the unpublished map and report of the Turkana area. We also thank Dr. P. S. Sahasrabudhe and Dr. C. R. K. Murthy for facilities offered to one of us (P.K.S.R.) to carry out thermal demagnetization in the Tata Institute of Fundamental Research, India. We thank Professor A. N. Hunter for initiating this work and for his constant support.

This work was made possible by a grant from the Rockefeller Foundation and by assistance with equipment given by Professor P. M. S. Blackett.

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(Manuscript received June 7, 1965;
revised October 21, 1965.)

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